

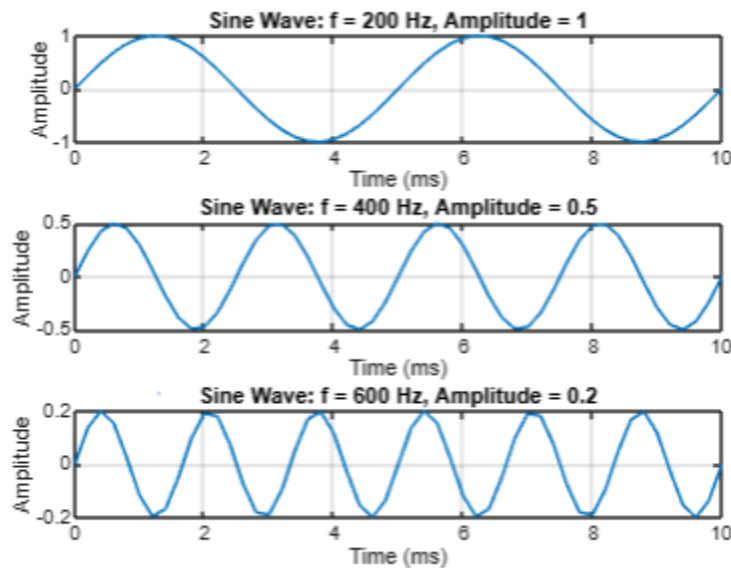
Task 1

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A.

How do different frequencies and amplitudes affect the shape of the waveform?

This is the plot output:



In the generated plots, we observe how varying the frequency and amplitude of sinusoidal signals affects their waveform shape.

- **Amplitude** controls the vertical stretch of the waveform. A higher amplitude results in taller peaks and deeper troughs. For example, the sine wave with amplitude 1 reaches higher and lower values than those with amplitudes 0.5 and 0.2, respectively. However, amplitude does not affect how frequently the wave oscillates.
- **Frequency** determines how many cycles occur per second. A higher frequency results in more oscillations within the same time interval, making the waveform appear more compressed horizontally. In the plots, the 200 Hz signal completes about 2 cycles in 10 milliseconds, while the 400 Hz and 600 Hz signals complete about 4 and 6 cycles, respectively.

In summary, amplitude affects the **height** of the waveform, while frequency affects its **density** (oscillation rate) over time.

B.

Describe the perceptual (auditory) differences between the three sine waves.

When played using MATLAB's `sound()` function, the three sine waves exhibit noticeable differences in pitch and loudness:

- The **200 Hz** sine wave sounds **low-pitched** and **loudest**, due to its lower frequency and highest amplitude.
- The **400 Hz** wave has a **moderate pitch** and is **less loud**, reflecting its mid-range frequency and reduced amplitude.
- The **600 Hz** signal is **high-pitched** and **the quietest**, with the highest frequency and smallest amplitude.

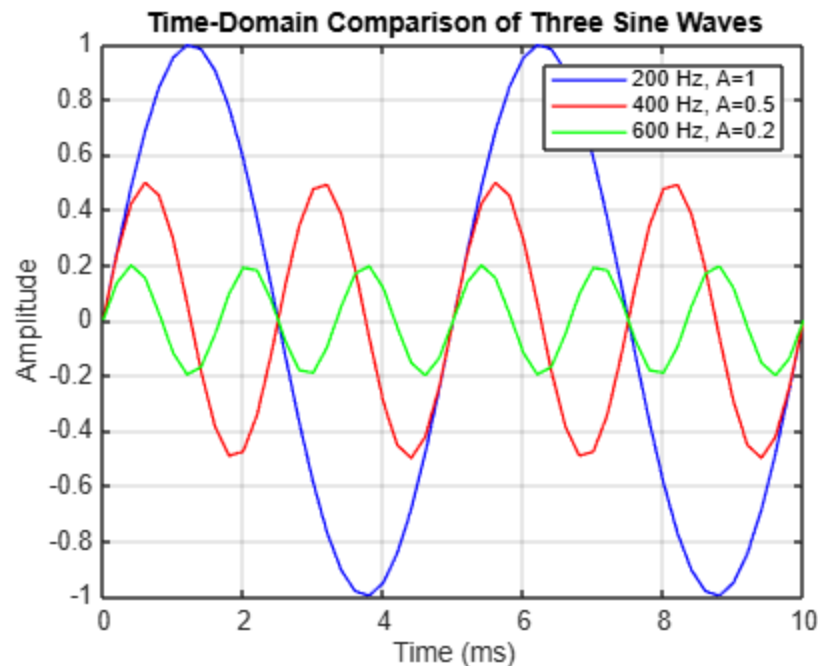
These perceptual characteristics align with human auditory perception:

- **Frequency** is perceived as **pitch** — higher frequencies sound higher in tone.
- **Amplitude** is perceived as **loudness** — greater amplitudes sound louder.

C.

How does the time-domain representation reflect differences in frequency and amplitude?

This is the plot output:



In the time-domain plot, amplitude differences are shown by the vertical height of each sine wave:

The 200 Hz wave has the largest peaks due to its amplitude of 1, while the 400 Hz and 600 Hz waves have progressively smaller peaks corresponding to amplitudes 0.5 and 0.2 respectively.

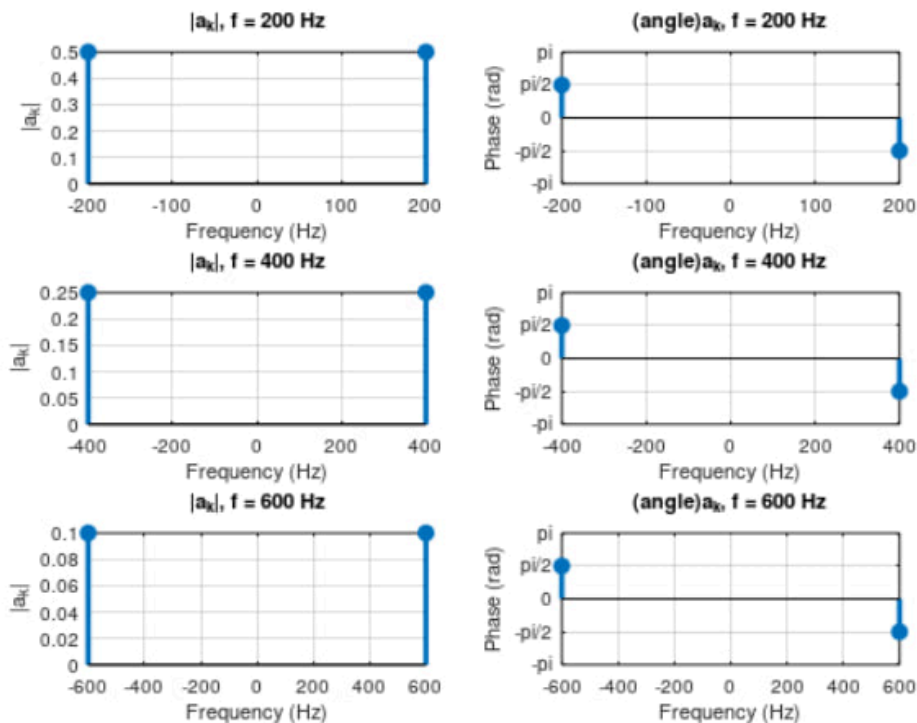
Frequency differences are reflected in how quickly each wave oscillates over time:

The 200 Hz wave completes fewer cycles within the time interval, resulting in widely spaced peaks, whereas the 400 Hz and 600 Hz waves oscillate faster, producing more cycles and closer peaks. Thus, amplitude controls the wave's vertical scale, and frequency controls the horizontal spacing of oscillations in the time-domain representation.

D.

How do the Fourier series plots compare to the original sine waveforms?

This is the plot output:



The Fourier series plots are the frequency domain breakup of the original sine waves. Since each original signal is a single frequency pure sine wave, its Fourier series contains only two non-zero coefficients at frequencies $+f$ and $-f$.

In the time domain, the sine wave shows continuous oscillations at frequency f .

- In the frequency domain, all the energy of the signal is in two discrete frequencies $+f$ and $-f$, which are the positive and negative frequency parts.
- Fourier coefficients at these frequencies are of amplitudes $A/2$ (A being the amplitude) and phases $\pm \pi/2$ (due to the imaginary unit in the sine representation).
- No other components of frequency are present, and that's the reason why plots of magnitude show two distinct spikes whereas phase plots show their respective angles.
- This perfect match guarantees that the Fourier series accurately and effectively approximates the provided original sine waveforms based on their fundamental frequency components alone.